

Allen Hsieh

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Education

University of California, Berkeley Expected May 2030
B.S. Aerospace Engineering, College of Engineering · Berkeley, CA

Troy High School May 2026
Troy Tech Magnet Program · Fullerton, CA

Experience

Systems Engineering Intern Summer 2025
Raytheon · Fullerton, CA

- Reverse-engineered an Arduino-based robotic system and produced its technical documentation, wiring diagrams, and CAD models.
- Shadowed engineering workflows across the mechanical, electrical, software, and systems departments; attended product development briefings and radar test site visits.

AFRL Scholars Intern Summer 2024
Air Force Research Laboratory · Albuquerque, NM

- Designed, built, and iteratively tested a staged-recovery rocket through 11 design iterations in OpenRocket and CAD; it reached 3,200 ft (barometric altimeter) and deployed its parachute recovery system.
- Operated FDM and resin 3D printers, a CNC lathe and mill, and a laser cutter to prototype solutions for autonomous robotics projects.
- Observed cold-atom laboratory research on inertial navigation systems based on Bose-Einstein condensates.

Logistics Director Fall 2026 – Present
Hackathons @ Berkeley · Berkeley, CA

- Plan catering and hacker bus transportation for Cal Hacks 13.0 (3,000+ attendees), and source event merchandise end to end, from vendor quotes to delivery.

Avionics Team Member Fall 2026 – Present
Cal Aero SAE · Berkeley, CA

- Built a universal motor test jig that measures thrust to guide motor selection for the competition RC plane.
- Designing an ESP32 telemetry and black-box system for autonomous turbulence correction through propeller power control.

Hacker & Attendee Summer 2026
Y Combinator Startup School · San Francisco, CA

- Selected to attend YC's in-person conference for top student builders; heard from Sam Altman, Garry Tan, and Jensen Huang on AI, deep tech, and building startups.
- Placed 5th of 1,000+ competitors at the Corgi Hackathon with RunItBack, built in 12 hours, which turns a single photo into an interactive 3D environment for dementia reminiscence therapy.

Leadership

Team Captain & Instructor
CyberPatriot

Sep 2022 – May 2026

- Led the team to 2nd place at the CyberPatriot National Finals twice (2025 and 2026; 1,406 teams in CyberPatriot 17) and to 1st place in the Cisco Networking Challenge both years.
- National semifinalist runner-up in 2023 and 2024, competing all four years as the team's Windows specialist.
- Taught about 200 students over two years as a program instructor and Troy CyberCamp staff.

Co-Founder & Programmer
VEX Robotics

Jul 2023 – May 2026

- Co-founded the school's first V5RC team and scaled it to a 7-team program of about 60 students, with 9 national appearances.
- Programmed autonomous navigation in C++ using sensor fusion, calculus-based odometry, motion profiling, and Monte Carlo algorithms.
- Back-to-back National Division Finalist (2024, 2025); organized \$11K in fundraising through community partnerships and a robotics camp for 50+ students.

Founder & President
Troy Preparedness Club

Jul 2023 – May 2026

- Founded a school-wide emergency preparedness initiative; hosted 3 Teen CERT and 2 First Aid/BLS trainings, certifying about 150 students and engaging 300+ families.
- Organized 2,000+ volunteer hours; earned a U.S. Congressional Certificate of Recognition and a team lead role on the FEMA Youth Preparedness Council (Region 9).

Founder & Team Captain
American Rocketry Challenge

Jul 2023 – May 2026

- Founded and captained a competitive rocketry team; raised \$2,000 from business sponsors.

Skills

Programming & tools	Python, C++, TypeScript, Git, MySQL, MATLAB, LaTeX
Design & fabrication	SolidWorks (CAD and FEA), Fusion 360, OpenRocket, CNC machining, FDM and SLA printing
Certifications	GIAC GFACT (cybersecurity), SolidWorks CSWA, FEMA CERT Basic Training, Red Cross CPR/AED

Awards

- CyberPatriot National Finals, 2nd place, and Cisco Networking Challenge, 1st place · 2025 and 2026
- NCSF Scholar with Honors · About the top 1% of 45,000 applicants, 2023
- VEX Robotics National Championships, Division Finalist · 2024 and 2025
- National Merit Commended Scholar and AP Scholar with Distinction
- U.S. Congressional Certificate of Recognition, and VFW Citation and Medal